

WASP-43b

R-1128

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220429 · created 2026-07-30T22:46:59+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459698.7375 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.057
Transit depth (Rp/Rs)^2	3.76 +/- 2.22 [%]
Orbital Inclination (inc)	80.49 +/- 2.81
Airmass coefficient 1 (a1)	0.92 +/- 0.22
Airmass coefficient 2 (a2)	0.031 +/- 0.089
Scatter in the residuals of the lightcurve fit is	22.56 %
Best Comparison Star	#2 - [230, 223]
Optimal Aperture	3.56
Optimal Annulus	9.17
Transit Duration (day)	0.045 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.057 vs 0.1594 (deviation 0.59σ)
Duration fit vs published	0.045 d vs 0.0512 d (deviation 0.4σ)
Mid-time O–C	1.1 min
Depth SNR	3.3
Residual scatter	22.56 %

Light curve

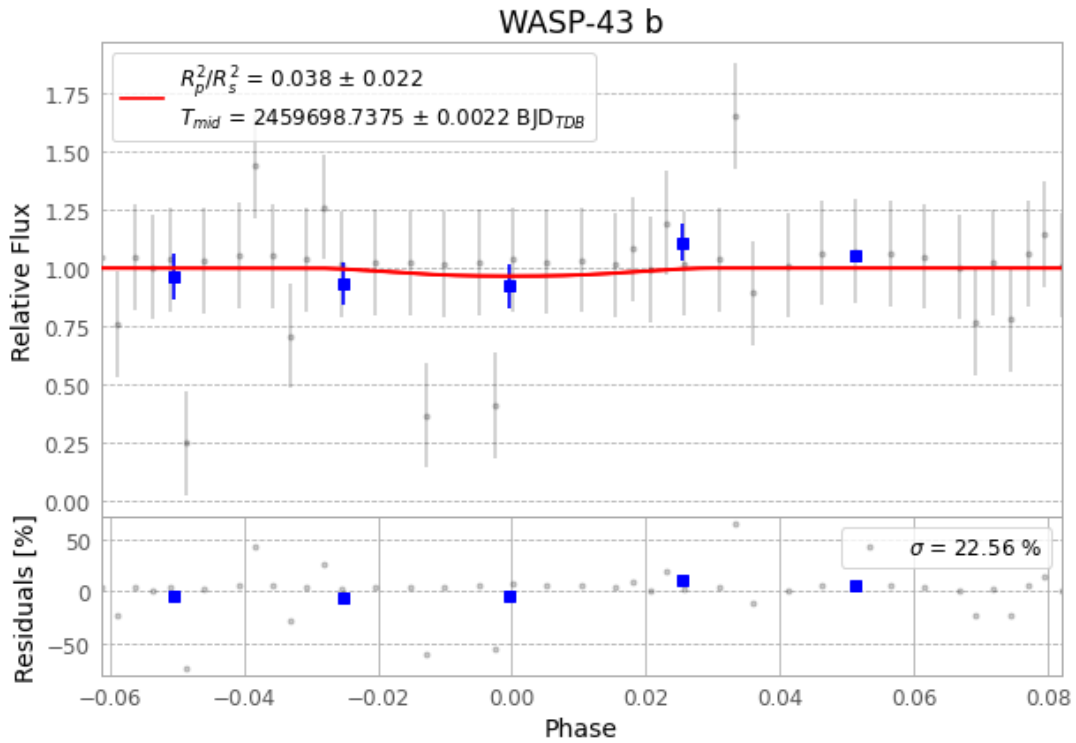


Figure 1 — FinalLightCurve_WASP-43 b_2022-04-28.png (SHA-256 2b95aa87b5d953cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 59c4ea978f78ffb9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

57 FITS frames (37 MB), WASP-43220429042412.FITS → WASP-43220429071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220429.log (c6b8c1421726...), FinalLightCurve_WASP-43 b_2022-04-28.png (2b95aa87b5d9...), FinalLightCurve_WASP-43 b_2022-04-28.csv (68b7a74b7241...)

Compute resources

Wall time 270.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 22:46Z.